Solution 1: Kullback-Leibler Divergence and model misspecification

The Kullback-Leibler Divergence is defined as:

$$\begin{aligned} D_{KL}(p \| q_{\boldsymbol{\theta}}) &= \mathbb{E}_{X \sim p}\left[\log\left(\frac{p(X)}{q(X|\boldsymbol{\theta})}\right)\right] \\ &= \underbrace{\mathbb{E}_{X \sim p}\left[\log(p(X))\right]}_{(a)} - \underbrace{\mathbb{E}_{X \sim p}\left[\log(q(X|\boldsymbol{\theta}))\right]}_{(b)} \end{aligned}$$

As we are looking for the set of parameters $\boldsymbol{\theta}$ that minimizes $D_{KL}(p \| q_{\boldsymbol{\theta}})$, we know the following:

- (a) does not depend on $\boldsymbol{\theta}$, and can be considered as a constant.
- To minimize $D_{KL}(p \| q_{\boldsymbol{\theta}})$ is equivalent to maximize (b)

Using the definition of the normal distribution:

$$\begin{aligned} (b) &= \mathbb{E}_{X \sim p}\left[\log(q(X|\boldsymbol{\theta}))\right] \\ &= \mathbb{E}_{X \sim p}\left[\log\left(\frac{1}{\sqrt{\sigma^2 2\pi}}\right) - \frac{1}{2}\frac{(X-\mu)^2}{\sigma^2}\right] \\ &= \log\left(\frac{1}{\sqrt{\sigma^2 2\pi}}\right) - \mathbb{E}_{X \sim p}\left[\frac{1}{2}\frac{(X-\mu)^2}{\sigma^2}\right] \\ &= -\log\sqrt{\sigma^2 2\pi} - \underbrace{\mathbb{E}_{X \sim p}\left[\frac{1}{2}\frac{X^2 - 2X\mu + \mu^2}{\sigma^2}\right]}_{(c)} \end{aligned}$$

Solving the component (c) we get:

$$\begin{aligned} (c) &= \frac{1}{2\sigma^2}\underbrace{\mathbb{E}_{X \sim p}\left[X^2\right]}_{\mathsf{Var}(X) + \mathbb{E}_{X \sim p}[X]^2} - \frac{2\mu}{2\sigma^2}\mathbb{E}_{X \sim p}[X] + \frac{\mu^2}{2\sigma^2} \\ &= \frac{2\sigma_0^2 + \mu_0^2}{2\sigma^2} - \frac{\mu\mu_0}{\sigma^2} + \frac{\mu^2}{2\sigma^2} \end{aligned}$$

Plugging (c) into (b), we get the expression that we want to maximize:

$$(b) = -\log\sqrt{\sigma^2 2\pi} - \frac{2\sigma_0^2 + \mu_0^2}{2\sigma^2} + \frac{\mu\mu_0}{\sigma^2} - \frac{\mu^2}{2\sigma^2}$$

To maximize (b), we differentiate the expression with respect to each parameter. We also need to compute the second derivative to be sure that the point is a maximum.

First, we differentiate with respect to the mean parameter μ:

$$\frac{\partial(b)}{\partial\mu} = 0 - 0 + \frac{\mu_0}{\sigma^2} - \frac{\mu}{\sigma^2} \stackrel{!}{=} 0 \longrightarrow \mu_{opt} = \mu_0$$

This value of μ is a possible maximum, we check the second derivative:

$$\frac{\partial^2(b)}{\partial\mu^2} = -\frac{1}{\sigma^2} < 0$$

As the second derivative is less than 0 at any point, μ_{opt} maximizes (b) and minimizes the Kullback-Leibler divergence accordingly. We now differentiate with respect to the variance parameter σ^2:

$$\begin{aligned}
\frac{\partial(b)}{\partial\sigma^2} &= -\frac{1}{2\sigma^2} + \frac{2\sigma_0^2 + \mu_0^2}{2\sigma^4} - \frac{\mu\mu_0}{\sigma^4} + \frac{\mu^2}{2\sigma^4} \\
&= -\frac{1}{2\sigma^2} + \frac{2\sigma_0^2 + \mu_0^2 - 2\mu\mu_0 + \mu^2}{2\sigma^4} \\
&= -\frac{1}{2\sigma^2} + \frac{2\sigma_0^2 + (\mu-\mu_0)^2}{2\sigma^4} \overset{!}{=} 0 \longrightarrow \sigma_{opt}^2 = 2\sigma_0^2 + \underbrace{(\mu-\mu_0)^2}_{=0 \text{ if } \mu=\mu_{opt}}
\end{aligned}$$

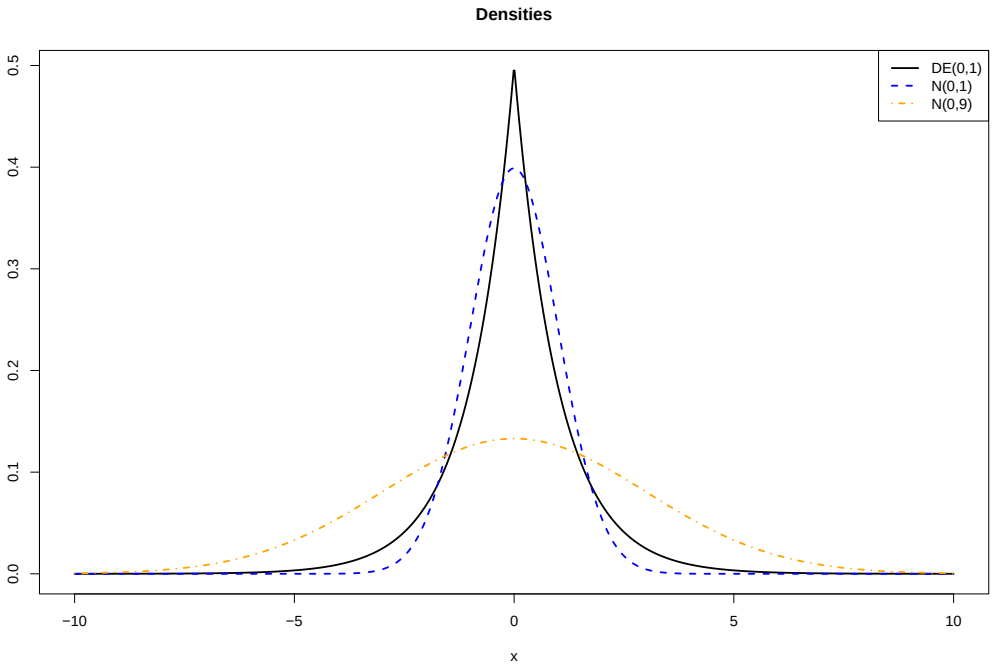
This value of σ^2 is a possible maximum, we check the second derivative:

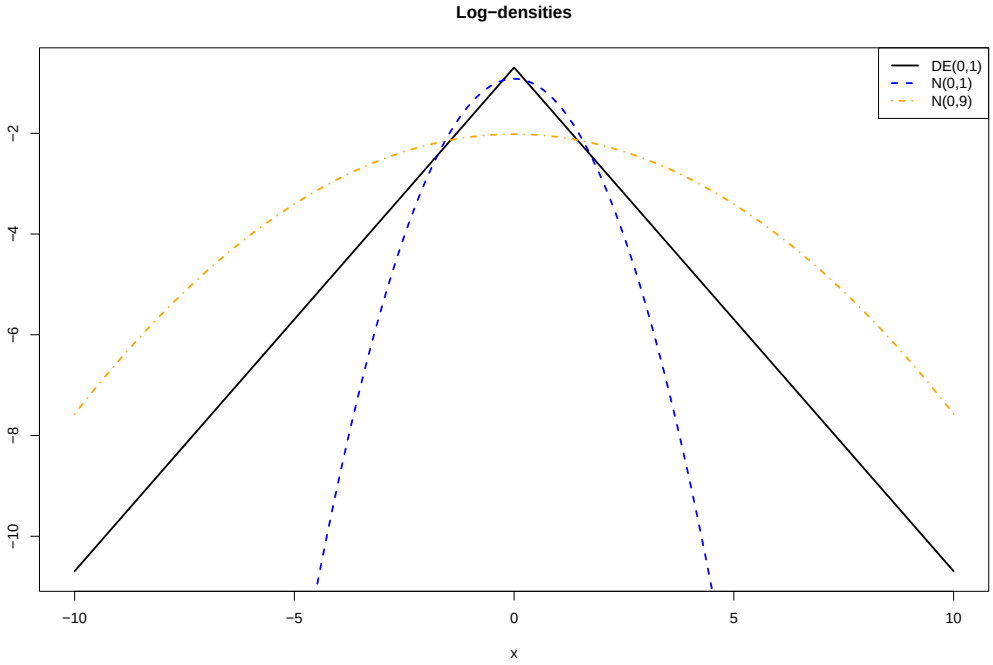
$$\begin{aligned}
\frac{\partial^2(b)}{\partial(\sigma^2)^2} &= \frac{1}{2\sigma^4} - \frac{2\sigma_0^2 + (\mu-\mu_0)^2}{\sigma^6} \\
\frac{\partial^2(b)}{\partial(\sigma^2)^2}\Big|_{\sigma^2=\sigma_{opt}^2} &= \frac{1}{2(2\sigma_0^2 + (\mu-\mu_0)^2)^2} - \frac{1}{(2\sigma_0^2 + (\mu-\mu_0)^2)^2} < 0
\end{aligned}$$

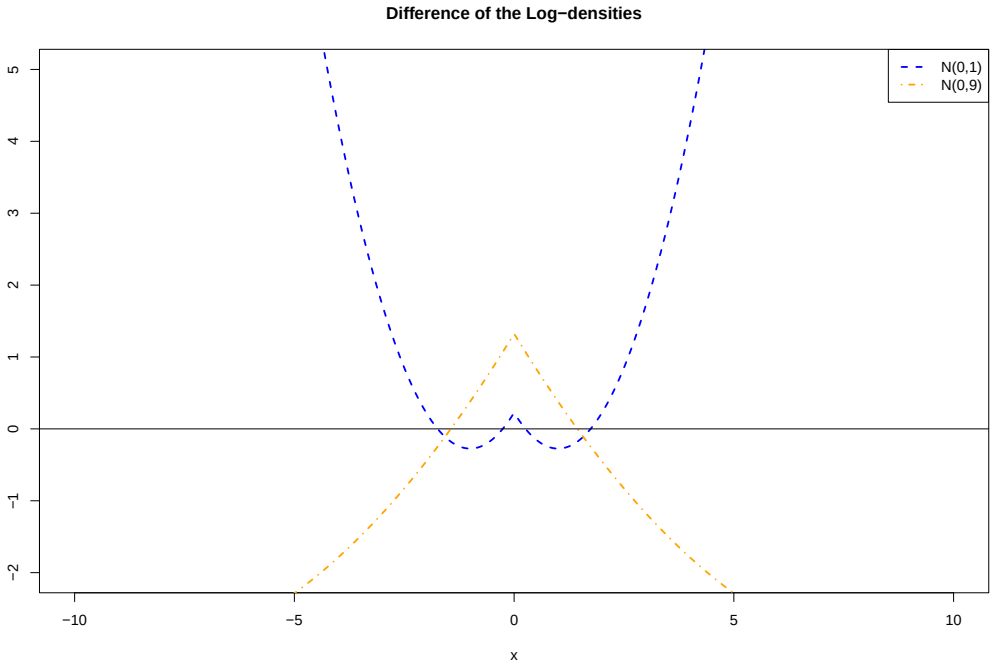
As the second derivative is less than 0 at the point we are looking, σ_{opt}^2 maximizes (b) and thus minimizes the Kullback-Leibler Divergence.

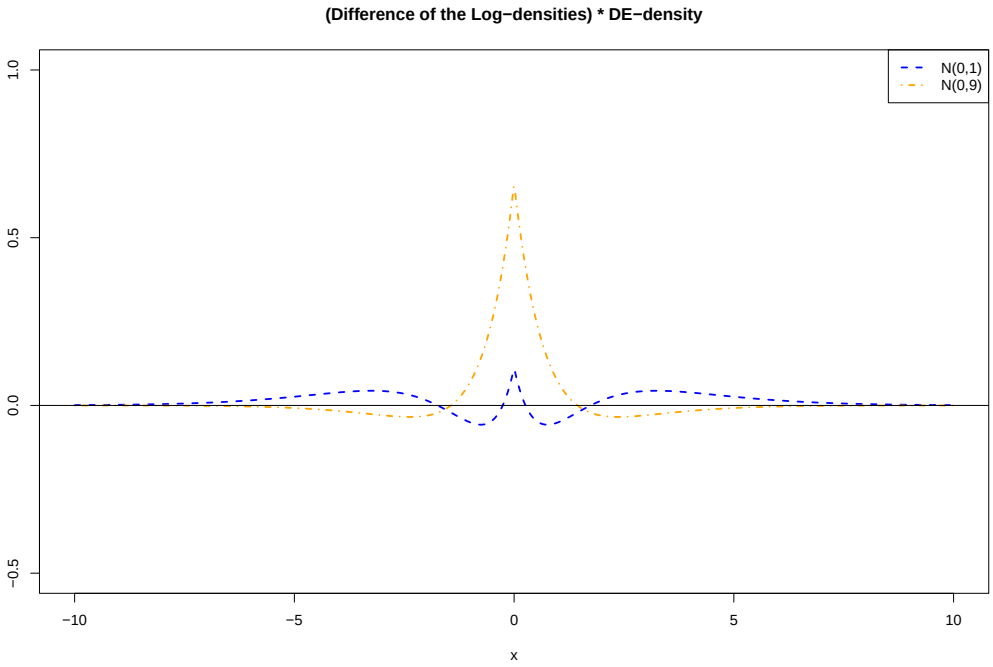
The following graphs may be helpful to understand the problem:

- The KL-Divergence, step by step:



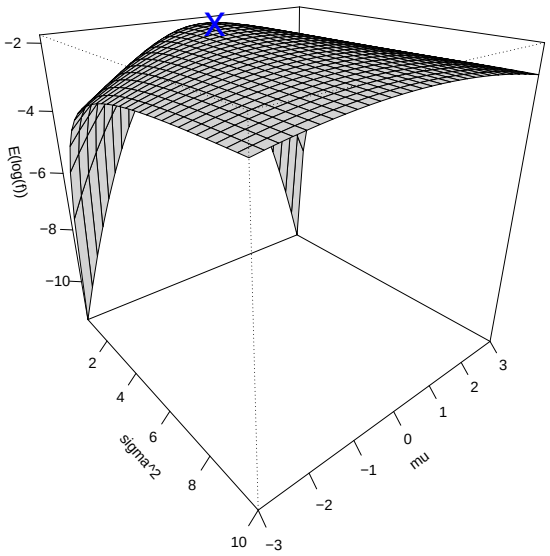

Log−densities
DE(0,1)
N(0,1)
N(0,9)
−2
−4
−6
−8
−10
−10
−5
0
5
10
x

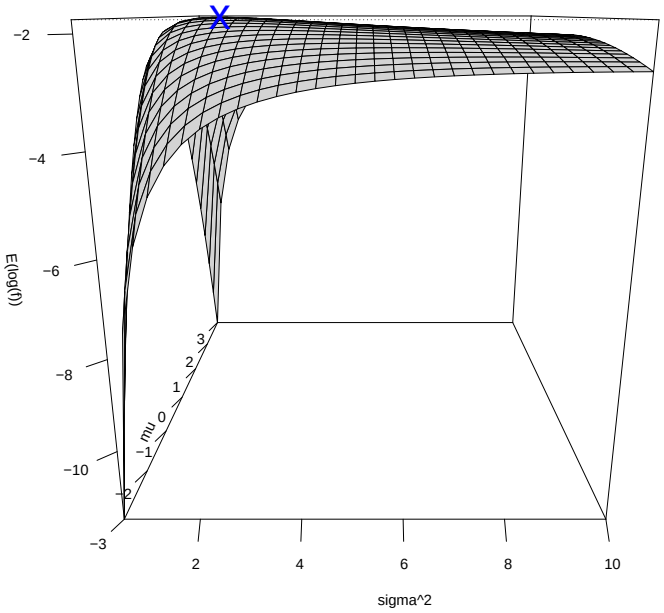

Difference of the Log−densities
N(0,1)
N(0,9)
5
4
3
2
1
0
−1
−2
−10
−5
0
5
10
x

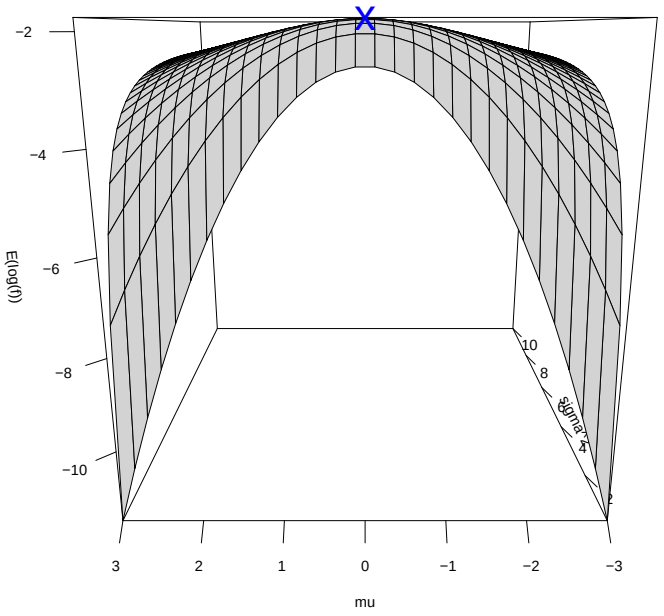


⇒ Now, the KLD is the integral of the function(s) in the last plot.

- 3D-Plot of the expression that we want to maximize (for parameters $\mu_0 = 0$ and $\sigma_0 = 1$):

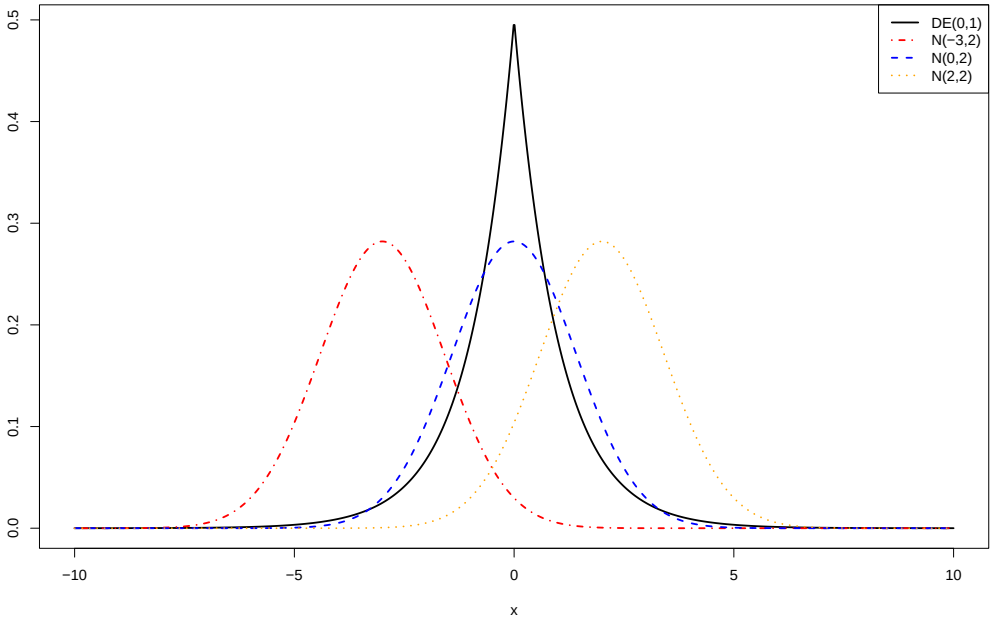






- Different configurations for the parameter μ and the optimal configuration:

Same Variance



Original and misspecified density, with optimal parameters.

